

Mohamed Mostafa

· Software Engineer ·

· Banha, Al Qalyubia · mohamed.mostafa0699@gmail.com · +20 114 231 4446 ·

· [LinkedIn](#) · [Portfolio](#) · [GitHub](#) ·

SUMMARY

Backend Engineer with 3+ years of experience building and maintaining production backend systems using Node.js and TypeScript. Experienced in designing scalable APIs, real-time services, background processing, cloud infrastructure, and payment integrations. Built and operated the backend for live marketplace platforms, focusing on performance, reliability, maintainability, and clean system design.

EXPERIENCE

Software Engineer II | [Lesoll](#) Jun 2023 – Jul 2026 · Cairo, Egypt · On-site

Worked as the sole Backend Engineer responsible for developing and maintaining the backend systems for **Lesoll** and **EgyStay**, managing production infrastructure, cloud services, and third-party integrations.

- Developed and maintained scalable REST APIs using **Node.js, Express.js, TypeScript, and MongoDB**.
- Built backend services for authentication, property management, bookings, payments, reporting, and admin operations.
- Implemented **Redis** caching and **BullMQ** background jobs to improve performance and reduce database load.
- Developed real-time features using **Socket.IO**, including chat, notifications, and online user tracking.
- Integrated **Paymob** and **Amazon Payment Services**, implementing secure payment workflows and webhook processing.
- Managed production infrastructure using **AWS (EC2, S3, CloudFront, Route 53), Docker, Nginx, PM2**, and **GitHub Actions**.
- Improved backend SEO by optimizing API performance and generating dynamic XML sitemaps for active listings.

PROJECTS

SRVJ — Real-Time Collaborative Whiteboard Platform [Live](#)

- Built real-time collaborative diagram editing using Yjs CRDTs over a custom WebSocket server implementing the Yjs sync and awareness protocols, enabling live cursors, presence, and seamless multiplayer editing.
- Secured WebSocket connections with PASETO v4 authentication and enforced project-level role-based access control, while isolating collaboration sessions into dedicated board rooms.
- Designed a dual-snapshot persistence layer that stores both the authoritative Yjs binary state for lossless recovery and a denormalized JSON representation for efficient querying and API reads.
- Architected the backend using PostgreSQL/Prisma for relational data (users, projects, memberships, sharing) and MongoDB/Mongoose for high-frequency diagram mutations.
- Implemented horizontally scalable notifications using SSE with Redis Pub/Sub fan-out, alongside BullMQ workers for asynchronous email and notification processing.
- Improved platform reliability and security through Zod validation, Helmet, CORS, rate limiting, centralized error handling, structured logging with Pino, and Prometheus metrics. Verified collaboration behavior with integration tests and containerized the platform using Docker Compose behind Nginx.

EDUCATION

Modern Academy – Maadi, Cairo, Jul 2022

Bachelor of Computer Science.

TECHNICAL SKILLS

Backend: Node.js, Express.js, TypeScript, RESTful APIs, WebSockets, SSE.

Databases & Caching: MongoDB, Mongoose, PostgreSQL, Prisma, Redis.

Messaging & Async: BullMQ, message queues, Redis pub/sub.

Payments: PayMob, Amazon Payment Services, Stripe (webhooks, HMAC, idempotency).

Cloud & DevOps: AWS (EC2, EKS, S3, ECR), Docker & Docker Compose, GitHub Actions CI/CD.

Frontend: Vue.js, Vite, TailwindCSS, and UnoCSS.